MicroBird: Embedded Bird-Audio Classification with Microjoule-Range CNN Inference on the MAX78002

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Abstract

Automated bird-audio classifiers can support passive acoustic monitoring, but many strong systems rely on cloud inference, global taxonomies, or watt-class edge hardware that is difficult to deploy at scale. This paper presents MicroBird, a region-specific 51-class bird-audio classifier deployed on the MAX78002 microcontroller CNN accelerator for Irish species monitoring. The dataset pipeline derives a regional species list from public occurrence sources, gathers focal recordings from Xeno-Canto, extracts 3-second clips, applies RMS gating and BirdNET teacher filtering, and converts accepted clips to 64×128 log-mel spectrograms. A compact hardware-aware CNN is trained with quantization-aware training (QAT) and synthesized using the ai8x toolchain. On a held-out Xeno-Canto-derived test split of 3,034 recording-ID-disjoint clips, the deployed end-to-end path achieves 79.10% top-1 accuracy, 90.90% top-3 accuracy, macro-F_1 of 0.792, and weighted-F_1 of 0.789. CNN inference latency is 3.5 ms; the full spectrogram-plus-CNN compute path requires 218.8 ms per 3-second clip. From measured durations and fixed power coefficients, CNN inference is estimated at 15.3 µJ and the full compute path at 2.168 mJ. The frontend accounts for approximately 99.3% of estimated compute energy, revealing how accelerator-backed classification shifts the system bottleneck from CNN inference to feature extraction. To the best of our knowledge, this is among the first reports of a deployed multi-class bird-audio classifier with microjoule-scale CNN inference on this hardware class. Evaluation is limited to held-out public-audio clips; long-duration field deployment remains future work.

Keywords: bioacoustics, embedded AI, bird classification, TinyML, MAX78002, passive acoustic monitoring, quantization-aware training

1 Introduction

Bird populations are widely used as indicators of ecosystem condition because changes in abundance and distribution reflect broader pressures such as land-use change, habitat degradation, and climate change [Gregory et al., 2005]. Passive acoustic monitoring extends this idea by allowing autonomous recorders to collect long-duration environmental audio without continuous human survey effort [Shonfield and Bayne, 2017, Hill et al., 2018]. For dense monitoring networks, raw audio alone is not enough: useful deployments also need local species-level screening that limits storage, bandwidth, and maintenance cost.

Large-scale bird-audio classifiers such as BirdNET show that spectrogram CNNs can recognise broad species sets with strong accuracy [Kahl et al., 2021]. However, many monitoring deployments do not require a global classifier; they require reliable screening of locally plausible species on hardware that can remain in the field. MicroBird addresses this narrower setting with a regional Irish species set and a classifier designed for deployment on a microcontroller-class device rather than a single-board computer.

MicroBird tests whether a regional class set and hardware-aware CNN can provide useful multi-class bird-audio accuracy at much lower embedded compute cost. The target hardware is the MAX78002, a microcontroller with an Arm Cortex-M4 host processor and a dedicated low-power CNN accelerator. The accelerator can execute quantized convolutional networks efficiently, but only when the architecture fits the ai8x synthesis flow, supported operators, memory layout, and layer scheduling constraints [Analog Devices, 2022a, Analog Devices, 2026b, Analog Devices, 2026a]. This makes quantization-aware training, spectrogram design, and firmware evaluation part of the same deployment problem rather than separate offline steps.

Accordingly, the evaluation considers model accuracy alongside stage-level latency and estimated compute energy to identify where optimisation effort has the greatest system-level effect.

This paper makes four contributions:

- A reproducible regional bird-audio dataset pipeline using public occurrence/audio sources and BirdNET-guided clip filtering.
- A 51-class hardware-aware spectrogram CNN trained with QAT and synthesized for the MAX78002.
- A deployed embedded pipeline with Cortex-M4 spectrogram generation and MAX78002 CNN accelerator inference.
- An empirical accuracy, latency, and duration-based energy evaluation showing useful multi-class accuracy and that frontend spectrogram computation dominates the deployed compute cost.

2 Related Work

Bird-audio recognition has shifted from hand-engineered acoustic features to spectrogram-based deep learning. BirdCLEF helped establish large-scale public evaluation for avian sound recognition, and by 2016 CNN-based spectrogram systems were already outperforming classical pipelines [Goëau et al., 2016, Sprengel et al., 2016]. BirdNET is the most relevant large-scale reference system: it uses deep spectrogram models to identify hundreds of North American and European species and has become a widely used tool for avian diversity monitoring [Kahl et al., 2021].

Xeno-Canto provides large numbers of community recordings with useful metadata, but labels are typically at recording level and recordings vary in duration, quality, background noise, and co-occurring species [Vellinga and Planqué, 2015]. The BirdSet benchmark paper highlights a broader reproducibility issue in avian bioacoustics: datasets, preprocessing choices, task definitions, and evaluation boundaries are often difficult to compare across studies [Rauch et al., 2025]. MicroBird therefore reports the regional species list, clip filtering rules, spectrogram settings, and recording-ID group split explicitly.

Low-power monitoring systems occupy a different design space from high-capacity server or cloud models. AudioMoth made low-cost acoustic logging practical, but it is primarily a recorder rather than a deployed multi-species classifier [Hill et al., 2018]. Bird@Edge moves recognition closer to the recording site using edge hardware and a distributed monitoring setup, but remains oriented toward more capable edge platforms rather than a microcontroller CNN accelerator [Höchst et al., 2022]. TinyChirp demonstrates that bird-song screening can run on an nRF52840-class microcontroller, but its reported task is binary corn-bunting detection rather than regional multi-species classification [Huang et al., 2024]. WrenNet is closest in spirit to MicroBird because it targets 70-species bird classification on low-power bioacoustic loggers and reports deployment on AudioMoth [Ciapponi et al., 2026]. MicroBird differs by using a dedicated microcontroller CNN accelerator, a 51-class Irish species set, and an explicit split between frontend spectrogram cost and CNN inference cost.

3 Method

3.1 Dataset Construction

The class set contains 50 Irish bird species plus `non_bird`. Candidate species were selected from eBird occurrence data and cross-referenced with BirdWatch Ireland monitoring resources [Sullivan et al., 2009, Lewis et al., 2020], then represented in JSON lists with numeric identifiers, common names, and scientific names. The pipeline is intended to scale to larger regional lists, but the deployed system evaluated here uses this 51-class task.

Recordings were collected from Xeno-Canto using species-level metadata manifests. Candidate recordings with quality ratings A–C were retained, species were capped at 350 recordings, and no more than 50 recordings per recordist were selected to reduce over-representation of a single recording style or environment. Recordings from Ireland, the UK, and nearby regions were preferred.

Each recording was split into non-overlapping 3-second clips after skipping the first second. An adaptive root-mean-square (RMS) gate discarded low-energy windows using the louder of -40 dBFS and the recording's

30th RMS percentile. Remaining windows were classified with BirdNET as a teacher. A window was accepted as the target species when BirdNET's top detection matched that species with confidence above 0.3. If no bird was detected above a 0.1 general bird threshold, the window was retained as `non_bird`; if another bird species was detected, the window was dropped. The `non_bird` class is therefore an initial rejection class derived from low-confidence public-recording windows, not a comprehensive environmental soundscape class.

The final spectrogram dataset contains 30,201 clips, split 80/10/10 into 24,147 train, 3,020 validation, and 3,034 test clips. Splitting was grouped by Xeno-Canto recording ID so that clips from the same source recording did not appear in multiple partitions.

3.2 Preprocessing

Audio was loaded as 16 kHz mono. Each 3-second clip was converted to a log-mel spectrogram using a 1024-sample short-time Fourier transform (STFT) window, hop length 256, 128 mel bins from 120 Hz to 8000 Hz, dB scaling relative to the clip maximum, and clamping to $[-80, 0]$ dB. Spectrograms were resized to 64×128 and normalised with a per-sample transform that subtracts the sample mean, divides by sample standard deviation, clamps to $[-3, 3]$, and rescales to $[-1, 1]$.

Training augmentations included SpecAugment-style time and frequency masking [Park et al., 2019], small time shifts, additive Gaussian noise, and mixup during floating-point pre-training only [Zhang et al., 2018]. Mixup was disabled during QAT so that quantization statistics were collected from real, unblended inputs.

3.3 Model, Training, and Deployment

A small architecture search selected the MicroBird model under the MAX78002/ai8x deployment constraints. Candidate models varied base channel width, additional convolution blocks, and residual connections while using the same split, preprocessing, class weighting, augmentation policy, and Adam training schedule. Validation accuracy improved substantially from 8- and 16-channel baselines to 32-channel models; adding moderate depth was more parameter-efficient than width alone at 16 channels, while residual variants did not improve QAT validation accuracy in this setting. The final model was selected because it provided the strongest balance of validation accuracy, synthesis compatibility, and deployment margin.

The MicroBird model is an ai8x-compatible CNN for a single-channel 64×128 spectrogram. It uses a 3×3 stem convolution, three fused max-pool-convolution stages, one additional non-residual 3×3 block, a 1×1 head convolution, fixed average pooling over the final 8×16 feature map, and a 51-output classifier. The synthesized model has seven accelerator layers, 393,160 parameters, and approximately 384 KB of 8-bit weights. Floating-point training used learning rate 10^{-3}; QAT began at epoch 30 with learning rate 10^{-4}, 8-bit weights, and best-checkpoint selection by validation loss. QAT is used here as a deployment requirement, not only as post-training compression, because integer arithmetic and ai8x layer constraints affect the deployable network.

The selected checkpoint was synthesized using ai8x and integrated into MAX78002 firmware. The Cortex-M4 computes the spectrogram frontend using CMSIS-DSP FFT routines, precomputed Hann windows and mel filters, dB scaling, resizing, int8 packing, and per-sample normalisation [Arm Ltd., 2025]. The CNN accelerator then executes the quantized network and returns top-k predictions. Figure 1 summarises this deployed path. Two evaluation paths were supported: `LOAD_PCM`, which uploads 3-second pulse-code modulation (PCM) audio and exercises the full frontend-plus-CNN path, and `LOAD_SPEC`, which bypasses spectrogram generation for debugging.

3.4 Evaluation

Final accuracy was measured by replaying the full held-out test split through the deployed UART/API evaluation path. Each 3-second PCM audio clip was sent to the board over UART; the firmware then ran the Cortex-M4 spectrogram frontend, executed the CNN accelerator, and returned top-k predictions. The host API was used only for replay and collection, so host-side transfer and control overheads are excluded from the

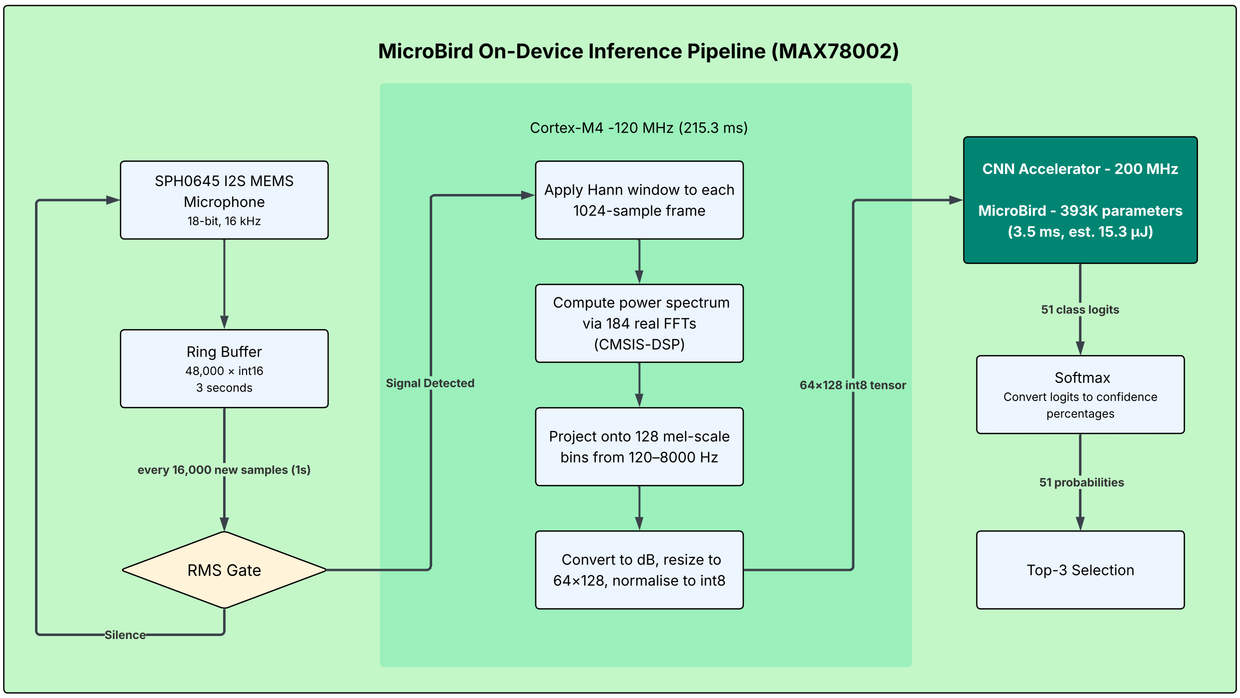
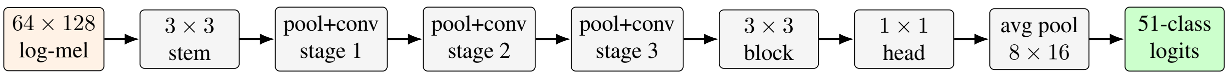


Figure 1: MicroBird deployed inference path. The Cortex-M4 computes the 64×128 log-mel frontend from 3-second audio, while the MAX78002 CNN accelerator runs the quantized classifier and returns top-k predictions. Evaluation replays equivalent 3-second PCM clips over UART.



Seven ai8x accelerator layers   393,160 parameters   ≈384 KB int8 weights

Figure 2: MicroBird CNN architecture synthesized for the MAX78002. The deployed network contains seven accelerator layers, 393,160 parameters, and approximately 384 KB of 8-bit weights.

embedded latency and energy figures. CNN latency was measured around accelerator execution, and spectrogram frontend latency was measured around the Cortex-M4 preprocessing path. The firmware also exposes MAX78002EVKIT PMON trigger windows for system and CNN stages following the vendor benchmarking flow [Analog Devices, 2022b]. Raw PMON serial logs were captured for representative validation runs, but a verified scalar energy value was not obtained from the PMON stream for all reported stages. Consequently, the scalar energy values reported below remain duration-based estimates from measured stage durations and fixed power coefficients. They include spectrogram frontend compute and CNN accelerator inference only, excluding microphone duty cycle, storage, radio transmission, host-side upload overhead, display, and idle-system energy.

4 Results

4.1 Deployed Accuracy

The deployed system achieved 79.10% top-1 and 90.90% top-3 accuracy on the held-out Xeno-Canto-derived test split (Table 1). Macro-F_1 and weighted-F_1 are close, indicating that the result is not driven only by the most frequent classes.

Table 1: Final deployed MicroBird performance on the held-out test split.

Metric	Value
Test clips	3,034
Classes	51
Top-1 accuracy	79.10%
Top-3 accuracy	90.90%
Macro-F_1	0.792
Weighted-F_1	0.789

Per-class results show that errors are concentrated rather than uniform (Table 2). Acoustically distinctive species such as sparrowhawk, kingfisher, buzzard, swift, and stonechat exceed 96% recall. The weakest class is `non_bird`, with 23.33% recall and 29.58% precision, confirming that it should be treated as an initial rejection class rather than a general environmental sound class.

Table 2: Best and worst deployed per-class recall values.

Class	Common name	Recall	n
Best classes			
Accipiter nisus	Sparrowhawk	100.00%	23
Alcedo atthis	Kingfisher	97.06%	34
Buteo buteo	Common Buzzard	96.92%	65
Apus apus	Common Swift	96.36%	55
Saxicola rubicola	Stonechat	96.05%	76
Worst classes			
`non_bird`	Initial rejection	23.33%	90
Prunella modularis	Dunnock	56.16%	73
Emberiza schoeniclus	Reed Bunting	56.86%	51
Anthus pratensis	Meadow Pipit	57.78%	45
Turdus philomelos	Song Thrush	58.18%	55

Table 3: Most frequent bird-to-bird confusion pairs in the deployed test run.

True class	Predicted class	Count
Dunnock	Kingfisher	10
House Sparrow	House Martin	9
Reed Bunting	Yellowhammer	7
Rook	Hooded Crow	6
Wood Pigeon	Collared Dove	6

4.2 Latency and Duration-Based Energy

Table 4 separates the measured latency and estimated energy of the frontend and CNN stages. CNN inference is cheap at 3.5 ms and an estimated 15.3 µJ. The full spectrogram-plus-CNN compute path is 218.8 ms and an estimated 2.168 mJ per 3-second clip. The frontend accounts for approximately 99.3% of the estimated spec-plus-CNN energy, indicating that spectrogram generation, rather than CNN inference, is the dominant optimisation target. This marks a system-level bottleneck shift in which accelerator-backed classification is no longer the dominant cost, making frontend optimisation more consequential than further CNN compression.

Table 4: Measured latency and duration-based energy estimates. Raw PMON trigger logs were captured for audit, but the scalar energy values here are estimated from measured duration and fixed coefficients.

Stage	Latency	Energy type	Energy
Spectrogram Cortex-M4	215.3 ms	Duration-based estimate	2.153 mJ
CNN accelerator	3.5 ms	Duration-based estimate	15.3 μJ
Total spec+CNN	218.8 ms	Duration-based estimate	2.168 mJ

4.3 Indicative Prior-Work Comparison

Table 5 compares MicroBird with BirdNET [Kahl et al., 2021], TinyChirp [Huang et al., 2024], and WrenNet [Ciapponi et al., 2026]. The comparison is indicative and not benchmark-equivalent because tasks, class sets, clip lengths, hardware, and measurement boundaries differ. MicroBird's energy value includes spectrogram frontend compute and CNN inference, while the CNN-only estimate is 15.3 μJ.

Table 5: Indicative comparison with related bird-audio systems. Comparisons are not benchmark-equivalent because tasks, metrics, clip lengths, hardware, and measurement boundaries differ.

System	Task	Hardware	Reported result	Energy boundary	Energy/clip
BirdNET	984 spp.	RPi 3B+	Global reference model	Frontend incl.; WrenNet bench.	2,790 mJ
TinyChirp	Binary	nRF52840	Precision >0.98; corn bunting detection	Frontend incl.; reported	42.5 mJ
WrenNet	70 spp.	AudioMoth	70.1% full-task accuracy	Frontend incl.; reported	77 mJ
MicroBird	51 classes	MAX78002	79.10% top-1; 90.90% top-3	Frontend incl.; estimate	2.168 mJ

5 Limitations and Future Work

The main limitation is evaluation scope. A held-out focal-recording test split is a common and useful stage for bird-audio model evaluation, and the split used here is recording-ID-disjoint. However, it is still derived from Xeno-Canto clips rather than from a long-duration outdoor field deployment. The reported accuracy is therefore evidence for deployment feasibility on the target hardware, not a final estimate of long-term field performance under weather, microphone placement, seasonal change, and persistent background noise.

The `non_bird` class is weak and should be expanded with a richer environmental corpus before being treated as robust background rejection. The scalar energy values are duration-based estimates rather than independently parsed PMON energy measurements. Finally, prior-work comparisons are useful for positioning but not benchmark-equivalent.

Future work should prioritise richer non-bird data, long-duration field trials, recordist- or location-disjoint evaluation, fleet-level deployment studies, and frontend optimisation. Since the CNN is already a small fraction of the estimated compute energy, sliding-window reuse, partial STFT reuse, or lower-cost features may improve total system cost more than further shrinking the classifier. The same curation pipeline can also be used to test larger regional species lists under the same hardware constraints.

6 Conclusion

MicroBird demonstrates that a 51-class regional bird-audio classifier can be deployed end to end on the MAX78002 with useful held-out accuracy, microjoule-scale CNN inference estimate, millijoule-scale full compute estimate, and a clear bottleneck shift from classifier inference to spectrogram frontend computation. The result supports hardware-aware regional classification as a practical direction for low-power acoustic monitoring, while leaving field validation and stronger non-bird rejection as necessary next steps.

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